

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29/09/2026
Team ID	SWTID-2026-2508
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture Diagram: Streamlining IT Procurement

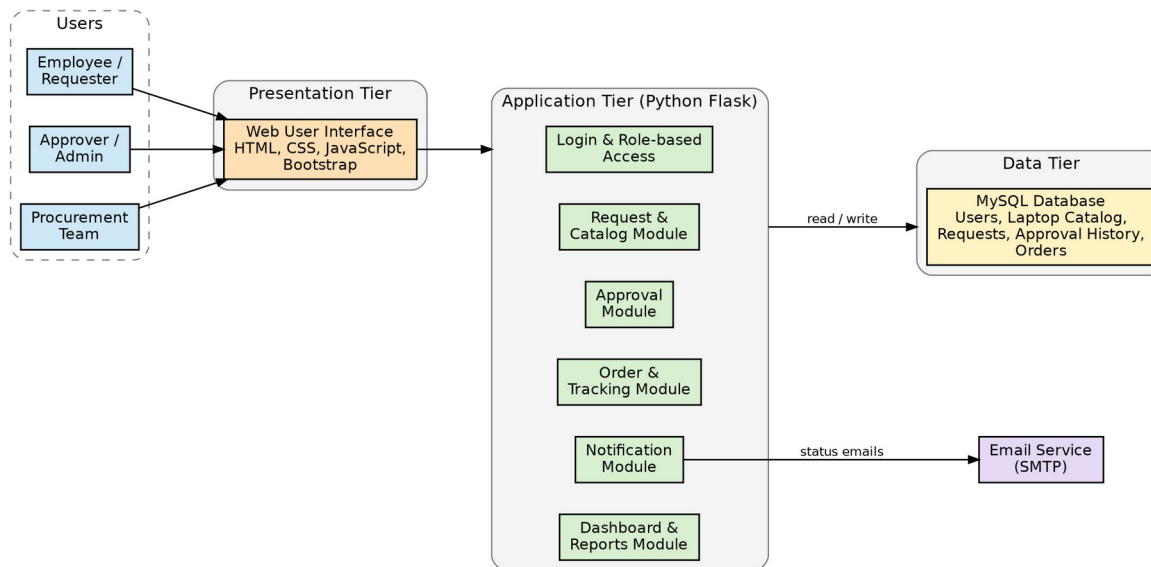


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web UI used by employees, admins and the procurement team to submit requests, approve them and track orders	HTML, CSS, JavaScript, Bootstrap
2.	Application Logic-1	Login, role-based access, laptop request submission and standard laptop catalog	Python (Flask)
3.	Application Logic-2	Approval workflow: pending requests, approve / reject with reason, approval history	Python (Flask)
4.	Application Logic-3	Order creation from approved requests, order status tracking, dashboard and reports	Python (Flask)
5.	Database	Stores users, standard laptop catalog, requests, approval history and orders as relational tables	MySQL
6.	Cloud Database	Not used in the current design; the database runs on the local server (can be moved to a cloud database later)	Not applicable (MySQL on local server)
7.	File Storage	Storage for exported reports and any uploaded documents	Local filesystem
8.	External API-1	Sends email notifications to the requester and admin when a request status changes	SMTP email service
9.	External API-2	Not used in the current design	Not applicable
10.	Machine Learning Model	Not used in this project	Not applicable
11.	Infrastructure (Server / Cloud)	Local server: Python 3, Flask with Gunicorn, MySQL. Cloud server: not used at present	Local server (Gunicorn / Flask, MySQL)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Flask (Python web framework), Bootstrap for the interface and MySQL as the database	Flask, Bootstrap, MySQL
2.	Security Implementations	Login with hashed passwords, role-based access control (employee, admin, procurement), session management, input validation and parameterised queries against SQL injection	Password hashing, RBAC, OWASP guidelines
3.	Scalable Architecture	3-tier architecture (presentation, application, database) with separate modules for request, approval, order, notification and reports, so each part can be extended or scaled separately	3-tier, modular Flask blueprints
4.	Availability	Application runs on a dedicated server with automatic restart and regular database backups so the system can be restored quickly after a failure	Gunicorn, scheduled MySQL backups
5.	Performance	Indexed database tables, paginated dashboard and request lists, and static files (CSS, JS) cached in the browser to keep pages fast	MySQL indexes, pagination, browser caching

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>